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Tailored Approach of Plug & Perf Completion Technique

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Abstract

Plug & Perf completion technique has been introduced with the aim of zonal selection flexibility which is required to deliver optimum multistage fracturing. This paper will cover a performance review for plug and perf operations for conventional reservoirs and capture several recommended techniques to enhance plug and perf performance efficiency with holistic evaluation of the operational improvement achieved.

The method starts by analyzing mass performance efficiency for number of plug & perf jobs performed and identifying the key metrics influencing performance efficiency. Then a set of practices were recommended and adopted to address the identified metrics with the aim of improving operational efficiency. These practices were evaluated based on a number of criteria: how long it takes to complete a fracturing stage, how often a fracturing stage requires contingent wireline/Coiled Tubing (CT) run, maintaining the zonal isolation while performing the frac operations & finally assuring stimulating most of the net pay. A set of operational enablers had to be put in place to raise efficiency of such type of completion to meet the targeted KPI's while meeting the completion objectives of wells. The study concluded several areas of improvement. Some of these areas of improvements were related to E-line operations such as E-Line loss of communication & plug failure. Also, some of these areas of improvement were formation related such as low injectivity stages, un-even treatment placement due to cluster depth & spacing selection. Moreover, additional areas of improvements related to other aspects were determined such as proximity of perforation to plug depth (plug integrity), liner deformation, difficulties to mill the plugs, cement integrity, and finally zonal communication. Solutions to tackle each of these shortcomings included wellbore loading, more frequent wireline cable re-heading, and adjustments to run-in-hole (RIH) speed. Other solutions included combining stages, cluster and plug depths selection criteria, utilizing re-engineered near wellbore bullheading diverters, introduction of slim & dissolvable plugs, and enhanced analysis of onsite pumping decision making. The outcomes for deployment of these practices resulted in a 68% reduction of operational time per frac stage with less frequent need for CT/wireline runs and a noticeable reduction of nonproductive time (NPT) exceeding 25%. All while assuring flawless treatments with isolated fractures propagating deep inside the reservoir and assured stimulating high percentage of the net pay.

This paper summarizes performance efficiency analysis for plug and perf operations. It provides insights into the best-recommended practices and optimization initiatives with a clear path to visualize the anticipated value gain from adopting such practices in plug and perf operations

Introduction

The plug-and-perf (PnP) completion technique has become the predominant method for multi-stage hydraulic fracturing in unconventional oil and gas reservoirs (1, 4). Its historical development mirrors the broader advancements in horizontal drilling, stimulation design, and the economic pursuit of hydrocarbons in low-permeability formations (1). Initially derived from simple cased-hole perforation and single-stage fracturing methods in vertical wells, plug-and-perf has evolved significantly to accommodate the complexity of modern unconventional field development (1).

In the pre-1990s era, hydraulic fracturing was largely confined to vertical wells in conventional formations. Completions were relatively straightforward, utilizing either open-hole or cased-hole designs with limited perforation intervals. Fracturing operations were typically executed in one or a few zones. However, the concept of perforating the casing and fracturing through the perforations commonly referred to as "perf and frac" laid the groundwork for more sophisticated techniques.

The transition to horizontal drilling in the late 1990s and early 2000s marked a turning point. As operators began targeting tight gas and shale plays such as the Barnett Shale (1), the need arose for a reliable method to isolate and stimulate multiple zones along long horizontal laterals (1, 4). Plug-and-perf emerged as a viable solution, involving the sequential perforation of specific clusters, setting bridge plugs to isolate treated zones, and repeating the process to complete the entire lateral in stages. This technique allowed for staged stimulation while maintaining the advantages of a fully cemented casing string.

Between 2005 and 2015, the widespread adoption of plug-and-perf in major areas was driven by multiple technological advancements (1, 4). These included the development of composite bridge plugs, more accurate wireline-conveyed perforation systems, and improved cementing techniques. The method's flexibility, cost efficiency, and compatibility with cemented casing made it a preferred choice over open-hole systems like ball-drop sleeves.

While plug-and-perf is best known for its role in shale and tight formations (4), it has also seen increasing use in conventional reservoirs, particularly in re-fracturing operations and multi-zone completions (4, 3). PnP allows for selective stimulation of previously untreated or underperforming zones. By isolating specific intervals within the same wellbore, operators can target zones with higher remaining hydrocarbon potential, optimizing well performance without the need for costly sidetracks or new wellbores.

In conventional reservoirs with moderate permeability, the use of plug-and-perf has been adapted to address water coning, heterogeneity, and compartmentalization (3, 6). Engineers have leveraged limited-entry perforation strategies, lower-rate treatments, and tailored fluid systems to avoid formation damage while still benefiting from the mechanical isolation that PnP offers. This adaptation has proven effective in offshore and onshore applications where zonal selectivity, operational efficiency, and cost containment is critical.

However, the adaptation of the plug-and-perf technique to conventional reservoirs has not been without significant areas of improvement. The inherent variability in reservoir properties, operational constraints, and wellbore conditions in these formations has led to a range of efficiency-limiting issues not commonly encountered in shale environments. These include complications related to tool reliability, mechanical integrity, and fracture placement effectiveness. As a result, conventional plug-and-perf operations often fall short of performance expectations when executed using designs and workflows optimized for unconventional reservoirs.

This paper addresses these encounters by presenting a detailed review of the most critical operational challenges observed during plug-and-perf completions that are applied for conventional reservoirs. Each issue is discussed through a series of numbered field cases, with corresponding solutions that were developed and implemented to overcome them. The discussion includes both procedural modifications and design adaptations, culminating in a tailored plug-and-perf methodology better suited to the characteristics

of conventional reservoirs. This customized approach not only improved operational efficiency but also enhanced the reliability and effectiveness of stimulation treatments across varied field conditions.

Enhancing Stage and Perforation Design through Integration of Reservoir Quality and Wellbore Integrity

Operational Context

Historically, stage placement and perforation design in plug-and-perf (PnP) completions particularly in conventional and mixed-reservoir systems have prioritized reservoir quality indicators such as porosity, permeability, and saturation (3, 6). However, this approach has shown limitations when applied to horizontal wells where wellbore conditions can vary significantly along the lateral. These variations are especially pronounced near the toe section, where drilling-induced damage, poor hole cleaning, and mud filtrate invasion commonly degrade both formation integrity and mechanical reliability.

Key Issues Encountered

In early field implementations, perforation clusters were selected based solely on petrophysical cutoffs derived from logs or core data, with little consideration given to mechanical feasibility or borehole quality. This led to challenges during fracture initiation, especially in the first few toe stages where mud damage and washouts were prevalent. Compounding the issue, plug setting locations were occasionally too close to perforation clusters, increasing the risk of casing integrity loss due to localized stress concentration or liner erosion factors not fully appreciated in early-stage design. Additionally, there was a reluctance to place perforations in mechanically favorable but petrophysically weak intervals, such as shale streaks or low-permeability layers, which could otherwise serve as effective fracture initiation points. By ignoring these zones, opportunities for stable initiation were missed, particularly in deviated or tortuous laterals.

Integrated Engineering Solutions

To address these limitations, operators transitioned to a more holistic approach:

1. **Mechanical and Reservoir Co-optimization:** Perforation and stage locations were selected based on an integrated analysis of mechanical stability, log-derived rock properties, and wellbore imaging, including caliper logs and borehole breakout data. This allowed for perforation placement in zones with adequate stress contrast and mechanical strength, even if reservoir quality was suboptimal, to ensure reliable fracture initiation.
2. **Revised Plug-Perforation Spacing Protocols:** A minimum standoff distance between plug seat locations and the nearest perforation cluster was standardized to avoid post-setting liner deformation or leakage due to shock loading and erosion effects.
3. **Deployment of CBL for Cement Verification:** Cement bond logs were used as a verification tool to screen for perforation zones with strong casing support, improving anchoring integrity during fracturing. Over time, running Cement Bond Logs (CBLs) on every well became increasingly cost-prohibitive. As a result, CBLs were selectively performed only on wells where the open hole caliper logs indicated abnormalities or potential zonal isolation issues. This targeted approach significantly reduced the number of CBL runs, improving overall operational efficiency. When CBLs were conducted, they were executed on the rig prior to transitioning into rigless operations. This timing ensured sufficient lead time to evaluate the CBL results and make informed decisions regarding the rigless intervention plan or to return the well directly for a work over operation.
4. **Contingency Planning for Stage Merging:** Where fracture initiation failed, a predefined fallback plan allowed operators to combine adjacent stages to ensure continuity in net pay stimulation while minimizing operational delays.

Outcome

The revised strategy led to a measurable improvement in both fracture initiation success rates and operational efficiency, with no significant loss in reservoir exposure or production performance. By incorporating mechanical reliability into stage planning, stakeholders were able to mitigate early-stage failure risks while maintaining productivity, even in challenging borehole environments.

Mitigating Tool and Plug Malfunctions by Accounting for Fluid Dynamics and Downhole Restrictions

Operational Context

Plug-and-perf completions require wireline-conveyed bottom hole assemblies (BHAs), which include plugs with tight tolerances relative to the casing inner diameter (11, 12, 13). In modern 5.5-inch casing strings, the clearance around composite or dissolvable plugs can be less than 0.25 inches per side. This minimal annular space creates a high-velocity bypass channel during pump-down operations, resulting in complex fluid dynamics that can produce suction zones, pressure drops, or swirling vortices around the plug.

Key Issues Encountered

In multiple field cases, deformed casing sections or internal restrictions such as doglegs and debris caused localized narrowing of the bypass area. This accelerated fluid velocities and reduced static pressure around the plug body, triggering rubber element swelling, premature plug setting, or more critically BHA pump-off due to tensile overload on the wireline weak point. In one case, repeated "hard tags" caused the plug to expand mechanically, jamming it against a restriction during a pull-out operation. Later surface valve manipulation inadvertently shifted downhole fluids, exerting unexpected hydraulic pressure and dislodging the wireline tool-string.

Integrated Engineering Solutions

1. **Pre-Job Simulation of Fluid Velocity Profiles:** Engineering teams began to simulate pump-down fluid velocities and plug behavior under various bypass conditions. This helped establish upper and lower thresholds for safe pump and line rates based on plug design and casing ID.
2. **Proactive Casing Integrity Screening:** Pre-run well integrity reviews were institutionalized, including caliper log analysis, CBL data, and historical drilling events to identify casing sections with high deformation risk.
3. **Revised operating procedures for Tool Movement Monitoring:** Updated procedures were created to monitor line tension spikes, vacuum signs, and differential fluid movement, which could serve as early indicators of bypass velocity anomalies or mechanical interference.
4. **Plug Design Modifications and Qualification Testing:** For high-risk wells, operators worked with plug vendors to validate element expansion limits, OD tolerance, and pressure differential resilience, tailoring designs to specific well geometries.

Outcome

By improving the understanding of hydrodynamic interactions around plugs and enhancing plug integrity screening, these practices significantly reduced the frequency of tool failures and enhanced overall job reliability. BHA pump-off events were minimized, leading to lower fishing risks and increased operational safety.

Customizing Dissolvable Plug and Fracture Design for Conventional Reservoir Application

Operational Context

Dissolvable plug technology has gained popularity in plug-and-perf operations due to its potential to eliminate mill-outs and reduce rig time (10). However, in low-temperature and low-salinity reservoirs, the dissolution rate of metallic and polymeric plug components can be significantly impaired. At the same time, poor fracture design alignment with formation stress profiles and mechanical boundaries has led to both premature plug failure and fracture containment issues, especially in gas-dominated or weakly consolidated formations.

Key Issues Encountered

Observed multiple completion failures where:

- Plugs skidded or lost sealing due to early dissolution under pressure.
- Incomplete dissolution caused by unfavorable downhole conditions required time-consuming drill-outs.
- Residual plug fragments migrated and formed pack-offs, delaying production handover.
- Unoptimized fracture azimuth and treatment pressure resulted in casing deformation and sanding, particularly in marginal gas plays.

Integrated Engineering Solutions

1. **Material Matching and Lab Testing:** Customized dissolvable plug materials were selected based on downhole temperature and salinity, using lab dissolution tests and field-calibrated dissolution curves.
2. **Fracturing Design Realignment:** Fracture modeling was applied, including mini-frac and DFIT tests, to validate fracture containment, azimuth targeting, and screen-out risk. Treatments were adjusted for stress contrast, modulus variation, and reservoir un-consolidation.
3. **Integrated Completion Workflow:** The entire completion sequence drilling azimuth, perforation phasing, plug design, and fracture schedule was executed under a unified engineering workflow with cross-discipline QA/QC checkpoints.
4. **Plug Supplier Collaboration:** Vendors were engaged early to simulate and qualify plug behavior in field-specific conditions, focusing on seal integrity, timed degradation, and pressure resilience.

Outcome

These measures resulted in improved fracture placement accuracy, minimized plug-related delays, and optimized completion economics in challenging gas and condensate reservoirs. By customizing plug and frac designs to local geomechanics, higher operational efficiency was achieved and improved EUR per well, especially in assets where mechanical risks had previously constrained performance.

Perforation Proximity to Isolation Plug

Operational Context

In modern well completions, particularly in multi-stage hydraulic fracturing with the plug and perf technique, isolation plugs are positioned strategically to isolate wellbore sections for individual stimulation treatments. Placing the perforation interval too close "less than 10 feet" to the isolation plug will introduce many operational and integrity concerns. A higher spacing is recommended between stages as it helps reduce the impact of casing deformation at the perforation points while maintaining adequate distance from the

frac plug that will be milled after fracturing treatments. Higher spacing between stages provides greater flexibility for setting a backup frac plug in the event of a failure of the primary plug. For fracture treatment that does not over flush at the end of pumping, this additional spacing also facilitates lower risk for wireline P&P operations. Proper understanding and managing of such risks are essential to maintain zonal isolation integrity and well productivity

Key Issues Encountered

If the perforation zone is very close to the isolation plug (less than 10 ft), several operational and integrity issues can arise. Those risks can be explained and categorized into several points:

1. Risk of Plug Damage:
The Detonation of perforating guns close to the isolation plugs creates powerful shock waves and pressure pulses. Such mechanical loads can move or deteriorate the plug if the spacing is too low. This can lead to zonal isolation failure and to potential communication between the zones and require remedial work.
2. Casing Deformation at perforation Points:
Perforating near isolation plugs can magnify casing deformation due to stress concentrations.
3. Cement or Formation Damage near Plug:
Perforation charges can also fracture or destabilize the cement or formation near the plug setting depth. This may affect the anchoring and stability of the plug and reduce long term reliability.
4. Operational Complications:
Close spacing between the plug and perforation zone can limit the tool-string placement and increase the risk during wireline or coiled tubing operations, especially during pressure tests or plug & perf operations.

Integrated Engineering Solutions

1. Maintain Sufficient Spacing:
Provide at least 10–15 feet spacing between the perforating area and the isolation plug to reduce the effects of shock waves and increase the accessibility of tools
2. Enhanced Cement Evaluation:
Perform cement bond logging before and after perforation to assess cement integrity, especially around plug seating zone
3. Optimized Perforation Design:
Carefully choose gun charges with controlled penetration and phasing to reduce near plug area damage while continuing to achieve good stimulation

Outcomes

Implementing these best practices reduces the risk of plug failure considerably, makes stimulation more efficient, and provides long-term zonal isolation. Diagnosis and re-design the perforation depths have improved reliability and reduced the number of remedial interventions

Guns Misfiring Frequency in Plug & Perf Operations

Operational Context

The performance of PnP has direct implications for reservoir access and production effectiveness (11, 12). Failure in this operation, specifically during perforating, can undermine zonal coverage, result in expensive remedial procedures, and contribute to non-productive time (NPT). Misfired guns can be caused by a lot

of factors which affect the overall operations and efficiency of the job and still it remains a significant operational challenge in the oil and gas industry.

Key Issues Encountered

Many root causes contribute to misfires in Plug and Perf operations; these include but are not limited to:

1. Electrical Failures:

Loose or incorrect wiring, particularly in wiring selective switches, is a major cause of misfires.

2. Mechanical Assembly issues:

Poor crimping, damaged connectors, or incorrect loading of perforation guns can disrupt the firing train

3. Environmental Impacts:

- High-pressure, high-temperature (HPHT) environments can degrade electronic components and cause insulation breakdown.
- Water or wellbore fluid entry due to poor sealing can shorten circuits.

4. Inadequate Quality Control:

- Manufacturing defects in explosive patches may lead to malfunctions in explosive components, & gun switches
- Inconsistent field assembly practices or failure to follow prescribed QA/QC protocols/checklist.

5. Switch Technology Limitations:

- Wiring Selective switches system: Prone to human error, restricted firing control, and increased misfire vulnerability (11, 12, 13).

Integrated Engineering Solutions

1. Standardized Field QA/QC Protocols/checklist:

- Including pre-run electrical continuity checks, mechanical inspections of tool-string & guns components, verification of switches and detonators integrity (to use Verification Box in some of explosive systems), checking the explosive patches, pressure and temperature rating compliance reviews to ensure system readiness before deployment.

2. Training and Standard Operating Procedures (SOPs):

- Maintain a very strict use to SOPs during the assembly, checks and execution of the operation to ensure precision in delivery of the job.

3. Advanced Switch Technologies:

- Addressable switches provide real-time downhole feedback, improving decision-making.
- Shifting from wiring selective switch to addressable selective system, which offers better error tolerance (i.e. less human error), more logging capabilities, and resistance to environmental conditions, more safety. In the following Table 1 shows a comparison that shows the main advantages & disadvantages of the two types of switches these are being used widely in the oil and gas industry:

Table 1—Comparison between wired selective systems and addressable selective systems

Feature	Wired Selective Systems	Addressable Selective Systems - Non-wired
Basic Principle	Physical wire connection to each gun (detonator & switch)	Unique address codes/IP for each Gun/Module without the need for wiring
Electrical Complexity & Connections	High. Requires numerous electrical connections at each gun joint and between gun strings. Many potential points of failure due to wiring errors, wires pinch points & damage	Low. Often features "plug-and-play" modular designs. Electrical connections are typically minimized, reducing human error during assembly and the likelihood of connection failures.
Reliability	Lower High susceptibility to misfires due to: wiring errors, poor or wet connections.	Higher. Significantly reduced misfire rates due to: minimized wiring, digital communication, advanced diagnostics, and enhanced safety protocols. Increased success rate per run.
Applicability	Still used in some simpler or less critical plug and perf operations	Dominant in multi-stage plug and perf for horizontal wells, complex completions, HPHT environments
Cost	Lower initial cost	Higher cost
Efficiency	Lower. More (NPT) due to misfires, required retrieval for troubleshooting/redressing, longer rig-up times, and the need to halt other operations (e.g., radio silence). Higher overall cost due to NPT.	Higher. Significantly reduces NPT by enabling bypassing of misfired guns, faster rig-up (plug-and-play), and often allowing simultaneous operations (no radio silence required). Leads to lower overall completion cost / stage.

Outcomes

The installation of engineering controls and switch technology upgrades has led to a considerable increase in perforating job reliability. Reports from the field showed:

- Up to 85% reduction in misfiring occurrences when converting from wiring to addressable configurations.
- Reduced rig-less time and less NPT to enhanced diagnostics and less unplanned interventions in the well.
- Enhanced well productivity by ensuring complete stage perforation.

There is a shifting trend toward the use of addressable selective switches in both high risk/profile jobs and normal profile. Although wired selective still remains in use for simpler/normal profile or budget-conscious projects, their long-term sustainability is diminishing due to ongoing reliability issues."

Evaluation

To quantify the impact of the tailored plug-and-perf strategy outlined in this study, a comparative analysis was conducted across 28 wellbores completed over a 12-month period. The evaluation was anchored on three key performance indicators (KPIs): average time per fracturing stage, non-productive time (NPT) incidence, and post-fracture integrity metrics (e.g., sustained zonal isolation and stage success rate).

Operational Time Efficiency

The average time required to complete a single fracturing stage dropped from 4.7 hours to 1.5 hours, representing a 68% reduction. This improvement was primarily driven by optimized plug setting protocols, enhanced wireline mobility through better RIH speed management, and avoidance of repeated tool runs due to BHA failures. These improvements translated into average savings of 27 hours per well, depending on the number of stages.

Reduction in Contingency Interventions

Prior to implementation of the enhanced approach, 31% of stages required unplanned coiled tubing or additional wireline runs due to either plug deployment issues, poor cluster isolation, or casing complications. After optimization, this figure dropped to 9%, signifying a 71% improvement. This reduction was achieved through more rigorous wellbore screening and real-time monitoring of wireline tension and pressure feedback.

Integrity and Treatment Success

The stage-level treatment success rate defined as the ability to initiate and sustain a fracture with clean fluid pressure response and without cross-stage communication rose from 82% to 97%. Post-fracture logs confirmed that cement bond optimization and revised perforation phasing played a significant role in maintaining zonal isolation and avoiding plug erosion.

Productivity and Early-Time Performance

Although the approach was designed primarily to improve operational reliability, early-time production analysis showed that the optimized completions yielded a 12–17% increase in average initial production (IP30) rates compared to offset wells using standard completion practices. This uplift is attributed to improved fracture placement accuracy and cleaner stimulation through enhanced near-wellbore treatment distribution.

Conclusions

The tailored plug-and-perf methodology developed and evaluated in this study demonstrates a clear operational and economic advantage when applied to conventional reservoirs, which are often underserved by shale-optimized workflows. By integrating reservoir mechanical considerations into perforation selection, improving wireline and plug logistics, and adapting fracture designs to local geomechanically constraints, significant gains were realized across all key metrics of performance.

The approach achieved:

- 68% reduction in per-stage completion time
- Over 70% decrease in contingency wireline/CT runs
- 15% increase in early production performance
- Marked improvement in operational reliability and zonal isolation success

These findings underscore the value of cross-disciplinary collaboration, including reservoir, stimulation, and well construction teams, to proactively address the unique complexities of conventional plays. The success of this field implementation presents a strong case for broader application of such tailored methodologies across similar assets globally.

Nomenclature

PnP	= Plug and Perf
CBL	= Cement Bond Log
BHA	= Bottom hole assemblies
SOP	= Standard Operating Procedure
NPT	= Non Productive Time
HPHT	= High Pressure High Temperature
QAQC	= Quality Assurance Quality Control

KPIs = Key Performance Indicators
RIH = Run In Hole
CT = Coiled Tubing
Net Pay = Portion of a Reservoir That Contains Economically Producing Hydrocarbons
CFD = Computational Fluid Dynamics

Subscripts

OD = outer diameter
ID = inner diameter
ft = feet

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